

Venkatesh P

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🌐 [linkedin.com/in/venkatesh-ml](https://www.linkedin.com/in/venkatesh-ml) 🐙 github.com/venkatesh-hyper 🌐 Portfolio

AI/ML Engineer specialising in **Retrieval-Augmented Generation, LLM Agents, and production MLOps**. Shipped a medical RAG system serving live queries on GCP (NASO Technologies) and a semantic search engine achieving **76,540 QPS** at **p95 = 0.07ms**. Experienced with the full GenAI stack: LangChain, FAISS, ChromaDB, HuggingFace Transformers, FastAPI, and Docker — precisely the toolchain required for building compliance intelligence systems at scale.

Technical Skills

GenAI / LLM	RAG Pipelines, LLM Agents (LangChain, LangGraph), Prompt Engineering, LoRA/QLoRA Fine-Tuning, RAGAS Evaluation, Sentence Transformers, HuggingFace Transformers, Groq API
Vector Search	FAISS (IVFFlat, HNSW, PQ), ChromaDB, Weaviate, Hybrid Search (BM25 + Dense), Reranking
ML / DL	PyTorch, TensorFlow, Scikit-learn, CNNs, RNNs, Ensemble Models, Transfer Learning
MLOps	Docker, Docker Compose, GitHub Actions (CI/CD), MLflow, DVC, FastAPI, Redis, Pydantic
Cloud	GCP (GCS, Vertex AI), AWS (EC2, S3), HuggingFace Spaces, Supabase
Data / Viz	Pandas, NumPy, Polars, PySpark, PostgreSQL, SQLite, Plotly, Power BI, Tableau
Languages	Python (advanced), SQL, JavaScript, C++

Experience

AI/ML Engineer Intern · *NASO Technologies*

Jan 2024 – Apr 2024

- Built **Well.AI** — a production medical RAG system processing real patient queries end-to-end: ingestion → LangChain chunking → HuggingFace embeddings → ChromaDB retrieval → LLM response; deployed via **FastAPI on GCP** with **Supabase** session logging for full query traceability
- Engineered prompt-engineering workflows for domain-specific health reasoning; structured retrieval context windows to improve response faithfulness for clinical question answering
- Containerised all services with **Docker Compose**; standardised multi-service deployment, eliminating environment-related bugs across dev/staging/prod
- Integrated LLM inference endpoints with GCP backend; implemented structured logging across all API calls enabling audit-ready traceability — directly transferable to regulated compliance environments

Machine Learning Researcher · *RUSA Project, University of Madras*

Apr 2024 – Jul 2024

- Engineered SNP-derived genomic features from GWAS data; trained ensemble ML models (XGBoost + Random Forest) achieving **87.6% accuracy** on Multiple Sclerosis risk prediction — published findings via internal research report
- Built Streamlit clinical decision-support dashboard for polygenic risk scoring, SNP visualisation, and model interpretability; deployed containerised workflow on **GCP with DVC** for full experiment reproducibility
- Demonstrated ability to translate complex domain requirements (genomics/clinical) into production ML systems — directly applicable to stakeholder collaboration in compliance engineering

Projects

PaperLens — High-Performance Semantic Search Engine

[GitHub](#) · [Live Demo](#)

sentence-transformers · FAISS IVFFlat · FastAPI · Groq LLaMA-3 · ChromaDB · Docker · Streamlit

- Indexed **956 arXiv papers** (scaling to 100K) via full pipeline: arXiv API ingestion → sentence-transformer embeddings (384-dim) → FAISS IVFFlat index (nlist=50, nprobe=10)
- Achieved **p50 = 0.02ms, p95 = 0.07ms query latency at 76,540 QPS** — production-grade retrieval throughput
- Built FastAPI backend with **LRU embedding cache**: reduced repeated query latency from 1,100ms to **0.44ms (1,738× speedup)**; exposed `/search`, `/summarize`, `/health` endpoints
- Integrated **Groq LLaMA-3.3-70b RAG pipeline** for structured 4-part paper summaries (~1,200ms); ChromaDB for chunk-level storage and retrieval

Ticket Analysis Bot — Enterprise RAG System

[GitHub](#)

FastAPI · ChromaDB · Groq Llama-4 Scout · SentenceTransformers · Streamlit

- Built production RAG pipeline over **70MB+ enterprise ticket logs**: PDF parsing → chunking → Sentence-Transformer embeddings → ChromaDB vector store → contextual QA via Groq Llama-4 Scout
- Achieved **p95 semantic retrieval latency under 200ms** via FastAPI REST; interactive Streamlit UI for natural-language querying across entire ticket dataset

- *Directly comparable to compliance document QA — same architecture applies to regulatory document analysis, audit log review, and obligation extraction*

Brain Tumor Classifier — CNN Medical Imaging

[GitHub](#)

TensorFlow · CNN · Transfer Learning · Docker

- Developed CNN achieving **95% accuracy** on 4-class MRI brain tumor classification (glioma, meningioma, pituitary, no tumor); containerised clinical inference interface with Docker

Education

Master of Computer Applications (MCA) · *University of Madras, Chennai* 2023 – 2025

GPA: 7.3 / 10 · Specialisation: Machine Learning, Deep Learning, Data Science

B.Sc. Biotechnology · *RKM Vivekananda College, Chennai* 2020 – 2023

GPA: 7.2 / 10 · Foundation in genomics, statistics, and computational biology

Achievements & Certifications

- **Winner** — AI Front-End Hackathon 2025, MCC Chennai
- **Finalist** — National AI Hackathon, Top 5% nationwide
- Published technical blogs on EDA, NLP pipelines, and Genomics ML (Medium)
- Vice President, Rotaract Club — led 5+ AI awareness and tech adoption community events
- **Certifications:** AWS Data Engineering Bootcamp (Udemy, 2024) · Machine Learning A–Z (Udemy, 2024) · Deep Learning Specialization (Coursera, In Progress)